Lecture 7 (Routing 4)

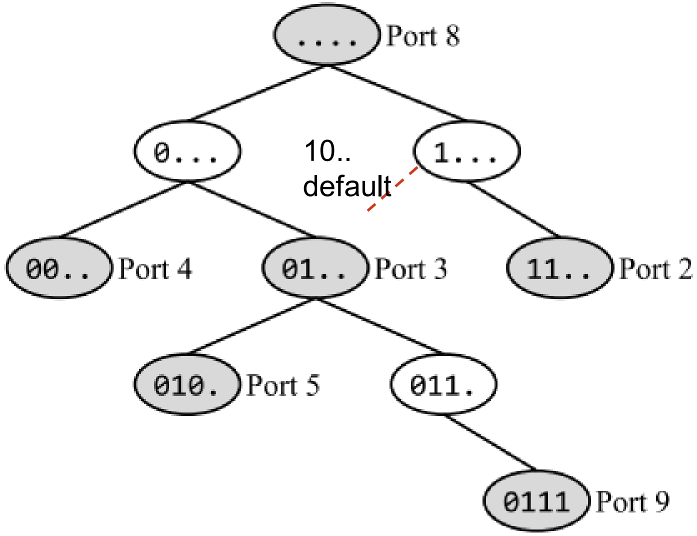
IP Routers Exercises ANS

Slides credit: CS 168 @ UC Berkeley

Q1. Longest Prefix Matching

In this question, consider the longest-prefix-matching trie below. All subparts are independent.

1) How many of the 16 possible 4-bit IP prefixes are forwarded along the default route?
2) In this subpart only, suppose we delete the node labeled 010. (Port 5). How many of the 16 possible 4-bit IP prefixes are forwarded along a different port using the modified trie (compared to the original trie)?
3) Which of these IP prefixes represents exactly the set of IPv4 addresses that will get forwarded along Port 2?
4) How many 32-bit IPv4 addresses are forwarded along Port 3?



Q1. Longest Prefix Matching ANS

1) How many of the 16 possible 4-bit IP prefixes are forwarded along the default route?

ANS: 4. All the 4-bit IP prefixes in the form 10.. will get forwarded along the default route, since it falls off the trie. There are 4 such prefixes: 1000, 1001, 1010, and 1011. The other 12 prefixes (with the form 00.. or 01.. or 11..) will not be forwarded along the default route

2) In this subpart only, suppose we delete the node labeled 010. (Port 5). How many of the 16 possible 4-bit IP prefixes are forwarded along a different port using the modified trie (compared to the original trie)?

ANS: 2. The prefix 010x, including 0100 and 0101, was forwarded along Port 5, but is now forwarded along Port 3. All other forwarding decisions remain unchanged.

3) Write the IP prefix that represents exactly the set of IPv4 addresses that will get forwarded along Port 2?

ANS: 192.0.0.0/2. IPv4 addresses are 32 bits, so the full prefix is: 11...... ....

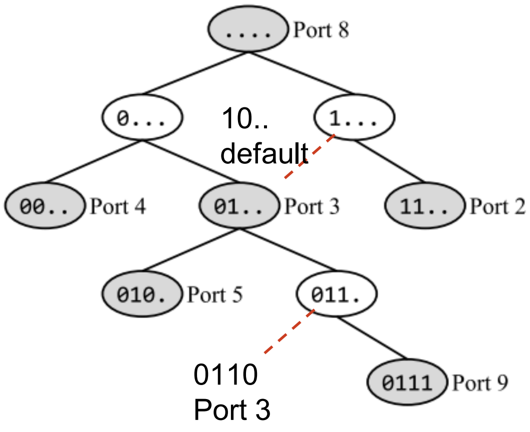
Filling in zeros in all the unset bits: 11000000 00000000 00000000 00000000, which is 192.0.0.0

Add /2 to indicate that first two bits are fixed: 192.0.0.0/2

4) How many 32-bit IPv4 addresses are forwarded along Port 3?

ANS: 2^{28}. Port 3 corresponds to the 01.. node. In this prefix, there are 30 bits unset, so there are 2^{30} IPv4 addresses in this range. However, we have to subtract all the addresses in the 010. range (29 bits unset, 2^{29} addresses), and the 0111 range (28 bits unset, 2^{28} addresses), since they get forwarded out of ports 5 and 9, respectively (based on longest-prefix matching). This gives us a total of $2^{30} - 2^{29} - 2^{28} = 2^{28}$ addresses forwarded along Port 3.

Another method is by realizing that only IPv4 addresses with 0110 as its prefix will be forwarded along Port 3 (based on longest-prefix matching). This means the next 28 bits can each be either 0 or 1, totalling to 2^{28} possible addresses



Red dashed lines are for illustration purposes, indicating what happens when falling off the trie. not part of the trie'.